

**From Ethical Principles to Executable Governance: Emotional Infrastructure Telemetry as
a Policy Translation Layer for Longitudinal AI Interaction Systems**

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Author Note

This manuscript presents Emotional Infrastructure Telemetry as a protocol-level governance architecture for AI systems that mediate longitudinal user interaction. The paper is framed as a conceptual standards and policy design contribution rather than an empirical validation study. Correspondence concerning this manuscript should be directed to Brittany Wright.

Abstract

Artificial intelligence governance often relies on high-level principles, post hoc audits, or documentation artifacts that do not directly control the longitudinal interaction pathways through which users experience AI systems. This paper develops Emotional Infrastructure Telemetry (EIT) as a policy translation layer for AI systems that must govern interaction continuity, consent boundaries, response escalation, and audit traceability across time. EIT is organized as a deterministic governance pipeline: Consent Scope, Longitudinal Interaction Logging (LIL), Trajectory Analysis and Response (TAR), Personal Development and Intervention (PDEV), and Audit Traceability. The contribution is protocol-level rather than diagnostic: EIT does not infer mental states, classify users psychologically, regulate model training data, or address compute-level frontier risk. Instead, it defines enforceable rules, schemas, control points, test vectors, and audit evidence for systems that adapt to user interaction histories. The paper integrates literature on AI risk management, algorithmic accountability, internal auditing, documentation practices, and socio-technical infrastructure, then translates that literature into deterministic logic suitable for conformance review. Appendices provide pseudocode, schema examples, an audit traceability matrix, and test scenarios. The central claim is that AI governance becomes operational only when abstract obligations are converted into observable system conditions, bounded permissions, auditable records, and enforceable escalation pathways.

Keywords: AI governance, algorithmic auditing, socio-technical infrastructure, telemetry, consent, auditability, longitudinal interaction, standards

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Introduction

AI governance has increasingly moved from abstract ethical declaration toward operational risk management, but a central translation problem remains unresolved. Regulators, standards bodies, platform operators, and civil society organizations can identify risks such as opacity, discrimination, manipulation, consent failure, and lack of accountability, yet many deployed AI systems still lack a protocol layer that makes these obligations executable during ordinary user interaction. The result is a gap between governance language and system behavior. Principles describe what should happen, but infrastructure determines what actually happens.

This paper addresses that gap by developing Emotional Infrastructure Telemetry as a policy translation layer for longitudinal AI interaction. The phrase emotional infrastructure is used here in a governance and systems sense, not as a claim that a system can diagnose, measure, or possess emotion. The concern is the interaction environment through which users encounter AI outputs, contest system behavior, receive explanations, preserve agency, and recover from breakdowns in continuity. EIT treats those interaction conditions as governance objects that can be logged, constrained, tested, and audited.

The paper builds from a core architecture consisting of Consent Scope, Longitudinal Interaction Logging, Trajectory Analysis and Response, Personal Development and Intervention, and Audit Traceability. Each component has a bounded function. Consent Scope determines whether the system has authority to process interaction context for a specific governance purpose. LIL records observable interaction events. TAR evaluates patterns across authorized records without making diagnostic claims about the user. PDEV selects proportional user-facing

responses. Audit Traceability records why each decision occurred, which rule governed it, and what evidence exists for review.

The contribution is intentionally protocol-level. A protocol-level governance paper must move beyond conceptual naming and provide deterministic logic, schema examples, testable rules, and traceability requirements. This paper therefore states the EIT model as a set of enforceable conditions. It also clarifies what EIT does not cover. EIT does not regulate model training data, compute allocation, model weight security, or frontier capability thresholds. Those domains require separate governance layers. EIT governs the interaction layer where deployed systems use, retain, escalate, or respond to longitudinal user context.

Research Problem and Scope

The core problem is that AI interaction governance is often evaluated at the wrong unit of analysis. A single model output can be inspected for correctness, harmfulness, or policy compliance, but many user harms and governance failures emerge across sequences of interaction rather than within a single output. A user may receive inconsistent explanations over time, encounter repeated refusals without meaningful recourse, experience hidden adaptation based on prior interactions, or lose the ability to distinguish whether a system is responding to the current prompt or to accumulated behavioral context. These are trajectory problems.

Trajectory problems require a governance architecture that can answer four questions. First, what interaction data was the system authorized to use? Second, what events were logged and for what purpose? Third, what pattern or trajectory justified a response change? Fourth, what audit evidence can prove that the system followed its policy obligations? EIT is designed to answer these questions without expanding the system into psychological profiling, behavioral targeting, or unbounded personalization.

The scope of EIT is limited to AI systems that maintain, evaluate, or respond to interaction context across time. The framework applies to chat interfaces, educational tutoring systems, workplace copilots, customer support agents, health information assistants, and other systems where longitudinal interaction can affect user agency, understanding, or access to escalation. The framework is not a general AI safety architecture. It is a governance layer for interaction authority, telemetry separation, user-facing intervention, and auditability.

Literature Review

AI governance and risk management

AI governance literature has increasingly recognized that trustworthy systems require institutional processes, risk controls, monitoring, and accountability rather than ethics statements alone. The NIST AI Risk Management Framework organizes AI risk work around governance, mapping, measurement, and management functions, creating a practical bridge between organizational obligation and risk control (National Institute of Standards and Technology [NIST], 2023). The OECD AI Principles similarly emphasize human-centered values, transparency, robustness, accountability, and inclusive growth as cross-jurisdictional norms (Organisation for Economic Co-operation and Development [OECD], 2024). The EU AI Act moves from principle toward enforceable obligations by imposing risk-tiered requirements for certain AI systems, including risk management, data governance, transparency, logging, human oversight, accuracy, robustness, and cybersecurity (European Parliament and Council of the European Union, 2024).

These governance approaches establish the necessity of risk management but do not fully specify the interaction-level mechanisms needed to govern longitudinal adaptation. EIT complements these frameworks by defining a narrower operational layer. It does not replace

organizational risk management or statutory compliance. It translates selected obligations, especially consent, transparency, logging, escalation, and traceability, into concrete protocol behavior at the deployed interaction layer.

Algorithmic accountability and internal auditing

Algorithmic accountability scholarship has shown that accountability requires more than disclosure of an algorithm's existence. Diakopoulos (2016) argued that algorithmic decision-making systems require scrutiny of inputs, outputs, goals, influence, and error. Kroll et al. (2017) developed legal and technical pathways for accountable algorithms, emphasizing that auditability must be designed into systems rather than reconstructed after harm occurs. Raji et al. (2020) advanced internal algorithmic auditing as an end-to-end organizational practice capable of exposing gaps between stated values and system behavior. Mökander et al. (2021) further positioned ethics-based auditing as an accountability mechanism for automated decision-making systems.

This literature supports EIT's central design choice: audit records must be generated as part of the system's ordinary operation. If audit evidence is created only after a complaint, the record will be incomplete, discretionary, and vulnerable to organizational self-protection. EIT therefore treats logging, rule matching, response selection, and integrity hashing as native governance functions. The audit record is not a separate report. It is a system artifact generated when policy logic is executed.

Documentation, model reporting, and data accountability

AI documentation practices have developed rapidly through model cards, datasheets, and related reporting formats. Model cards provide structured descriptions of model performance, intended use, limitations, and ethical considerations (Mitchell et al., 2019). Datasheets for

datasets provide structured context about dataset motivation, composition, collection, preprocessing, uses, distribution, and maintenance (Gebru et al., 2021). Suresh and Gutttag (2021) expanded this documentation logic by identifying sources of harm throughout the machine learning life cycle, including representation, measurement, learning, evaluation, deployment, and feedback loops.

These documentation practices are essential, but they are not equivalent to live governance of interaction trajectories. A model card can state that a system should not be used for a harmful purpose, and a datasheet can document training data limitations, but neither artifact necessarily records whether a deployed system used longitudinal context within the user's consent scope. EIT fills a different accountability position. It produces records of interaction-layer authority and response behavior while the system operates.

Socio-technical infrastructure

Socio-technical scholarship emphasizes that infrastructure is not neutral background machinery. Star and Ruhleder (1996) described infrastructure as relational, embedded, learned as part of membership, linked with conventions of practice, and visible upon breakdown. Bowker and Star (1999) showed how classification systems structure social life by determining what can be named, tracked, ignored, or institutionalized. Selbst et al. (2019) argued that fairness failures often arise when technical abstractions detach systems from their social context. Green (2021) likewise criticized approaches to technology ethics that fail to confront institutional power and political contestation.

EIT adopts this socio-technical view by treating interaction records as infrastructure rather than mere analytics. The system's authority to observe, retain, evaluate, and intervene shapes the user's practical agency. For that reason, telemetry must not be treated as a neutral

byproduct. It must be governed as infrastructure with purpose limits, access controls, audit obligations, and explicit non-use constraints.

Standards, conformance, and protocol logic

Standards governance requires more than persuasive language. The IETF convention for normative language, developed through RFC 2119 and clarified by RFC 8174, distinguishes mandatory, prohibited, recommended, discouraged, and optional requirements through terms such as MUST, MUST NOT, SHOULD, SHOULD NOT, and MAY (Bradner, 1997; Leiba, 2017). This convention matters because it converts prose into implementation obligations. ISO/IEC 42001 provides an organizational management-system frame for AI governance, while technical standards and machine-readable schemas define how specific controls can be implemented and tested (International Organization for Standardization, 2023).

EIT borrows this standards logic by defining policy as executable conditions. A policy is not complete when it states that a system should respect consent. It becomes operational only when the system knows what consent object must exist, what fields must be present, what action is blocked when consent is missing, what audit record is generated, and what review path is triggered when a boundary is contested.

Human agency, contestability, and rights protection

Governance literature also emphasizes the need to preserve human agency, contestability, and meaningful oversight. The GDPR's provisions on automated decision-making and data subject rights reflect a broader legal concern that individuals must have routes to understand, challenge, and constrain consequential processing (European Parliament and Council of the European Union, 2016). Floridi et al. (2018) framed responsible AI around beneficence, non-maleficence, autonomy, justice, and explicability. Jobin et al. (2019) showed that transparency,

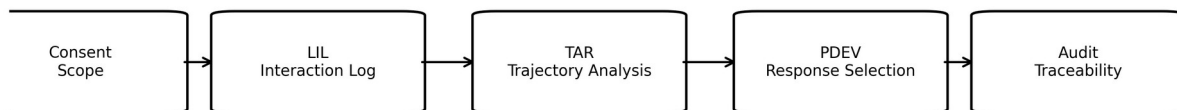
justice, non-maleficence, responsibility, and privacy recur across global AI ethics guidelines, even though implementation remains inconsistent.

EIT contributes to this implementation problem by creating a user-facing response layer. The PDEV component does not attempt to optimize the user's behavior. Its function is to preserve agency during governance-relevant interaction events by disclosing relevant system behavior, requesting clarification, pausing adaptation, routing to human review, or exporting records when appropriate. This converts rights language into system operations.

System Overview

EIT is a governance protocol for longitudinal AI interaction. Its system boundary begins when an AI application receives or generates an interaction event that may be logged for continuity, safety, accountability, or user-facing governance. Its system boundary ends when the event, any derived trajectory signal, any intervention response, and the relevant audit record have been created, retained, and made reviewable according to policy.

The pipeline is intentionally sequential. Consent Scope must be evaluated before any longitudinal telemetry is created. LIL records only authorized observable events. TAR evaluates authorized windows of LIL records and creates trajectory signals using explicit rules. PDEV selects a proportionate response based on the TAR signal and policy constraints. Audit Traceability binds each step to a reviewable record.



EIT governance pipeline: authority is checked before telemetry is created; every downstream action is linked to an audit record.

Figure 1. EIT governance pipeline from consent authorization through logging, trajectory analysis, response selection, and audit traceability.

Layered Architecture

Layer	Primary mechanisms	Governance function
Technical layer	Schemas, event records, rule engine, integrity hashes, access controls, API gates.	The technical layer converts governance obligations into machine-readable structure and deterministic execution paths.
Platform layer	Application interfaces, user consent surfaces, logging services, escalation channels, export tools.	The platform layer determines how users encounter the governance system and whether disclosures, pauses, review routes, and records are actually accessible.
Governance layer	Policy rules, conformance classes, test vectors, review ownership, retention limits, prohibited secondary uses.	The governance layer defines what the system is allowed to do, what it must not do, and how evidence of compliance is produced.
Human impact layer	User understanding, agency preservation, contestability, continuity, and recovery from breakdown.	The human impact layer evaluates whether the governance controls preserve practical agency rather than simply documenting system activity.

The layered architecture matters because control is distributed. A consent policy can be well written but ineffective if the platform interface hides the setting. A schema can be complete but harmful if the retention policy allows unbounded secondary use. An audit log can be technically valid but socially useless if no review owner is accountable for acting on it. EIT therefore treats governance as a layered control system, not as a single compliance document.

Control Points and Power Analysis

Control point	Power holder	Decision made	Failure or abuse scenario
Consent boundary	Platform operator and data controller	Whether longitudinal context may be logged or evaluated.	Hidden defaults, bundled consent, purpose creep.
Telemetry schema	System architect and standards implementer	Which events become visible to governance review.	Overcollection, undercollection, omission of contestation events.
Trajectory rule engine	Model/platform operator	Which patterns trigger disclosure, pause, review, or intervention.	Opaque scoring, inferred vulnerability, unreviewable thresholds.
Intervention interface	Application owner and UX designer	How users receive explanations, choices, appeals, and records.	Dark patterns, inaccessible notices, false choices.
Audit ownership	Compliance office, auditor, institutional review body	Who can inspect records and require correction.	Self-audit without independence, missing evidence, weak enforcement.

The control points reveal why EIT must include power analysis. A system that only logs events without governing who defines the log, who sees the log, and who can act on the log will not preserve accountability. The point is not to remove power from the system. The point is to locate power, bind it to explicit authority, and make its use reviewable.

Risk Map

Risk	Likelihood	Impact	Failure scenario	Mitigation pathway
Consent bypass	Medium	Critical	The system evaluates longitudinal context without a valid consent scope.	Block processing, generate audit exception, notify review owner, require consent repair before continuation.
Telemetry overreach	Medium	High	Interaction logs collect sensitive or irrelevant content beyond governance purpose.	Minimize fields, redact content where possible, enforce purpose-bound schemas, delete

Risk	Likelihood	Impact	Failure scenario	Mitigation pathway
				nonconforming records.
Diagnostic drift	Low to medium	Critical	Trajectory analysis becomes psychological classification or vulnerability scoring.	Prohibit diagnostic labels, restrict signals to observable governance events, require rule review for new signal types.
Hidden adaptation	Medium	High	The system changes behavior based on prior records without disclosure or user control.	Require adaptive disclosure, provide session boundary controls, audit response selection.
Audit manipulation	Low	Critical	Records are altered, deleted, or selectively generated after a complaint.	Use append-only logs, integrity hashes, independent review access, and exception alerts.
Escalation failure	Medium	High	The system detects a high-severity trajectory but does not route to human review or user-facing remedy.	Bind severity classes to mandatory response actions and test vectors.

The risk map is designed for conformance testing. Each risk is expressed as a scenario that can be simulated. This prevents the framework from becoming a list of values without enforcement mechanisms.

Policy Logic

EIT treats policy as executable logic. A policy rule is complete only when it identifies a condition, a required action, a prohibited action, an enforcement trigger, an escalation pathway, and an audit artifact. The following rules form the minimum policy logic for an EIT-compliant system.

Rule	Condition	Required system action	Escalation pathway
EIT-R1 Consent precondition	No valid consent_scope_id exists for longitudinal processing.	Do not create LIL record for longitudinal use; create consent	Consent repair required before

Rule	Condition	Required system action	Escalation pathway
		exception audit record.	processing resumes.
EIT-R2 Purpose limitation	The requested telemetry purpose is outside the consent scope.	Block processing and record purpose_mismatch.	Compliance review if repeated mismatch occurs.
EIT-R3 Mandatory observable logging	A governance-relevant event occurs within scope.	Create LIL event record with required fields.	System error if record cannot be created.
EIT-R4 No diagnostic inference	A rule attempts to assign psychological, medical, or vulnerability status.	Reject signal generation.	Policy review required for attempted prohibited signal.
EIT-R5 Trajectory window boundary	TAR evaluation is requested outside authorized window.	Limit evaluation to authorized records only.	Audit boundary reduction.
EIT-R6 Severity binding	TAR signal severity is high or critical.	Select required PDEV response type according to severity table.	Human review for critical severity.
EIT-R7 User-facing disclosure	System adaptation is based on prior interaction records.	Provide concise disclosure or explain the relevant context boundary.	Escalate if disclosure fails.
EIT-R8 Audit integrity	Any LIL, TAR, or PDEV record is created.	Create or update linked audit record with integrity hash.	Alert review owner if hash chain breaks.
EIT-R9 Non-use prohibition	Telemetry is requested for advertising, ranking, or behavioral targeting.	Deny request and create prohibited-use audit record.	Mandatory compliance review.
EIT-R10 Record export	User or authorized reviewer requests governance record access.	Export authorized records in readable and machine-readable format.	Appeal route if denied.
EIT-R11 Retention boundary	Record exceeds retention class without extension authority.	Delete or archive according to policy.	Retention exception audit required.
EIT-R12 Conformance evidence	System claims EIT conformance.	Pass schema, policy, and test-vector validation for claimed class.	Conformance claim must be withdrawn if validation fails.

The rules above are stated in implementation form. They are not merely institutional preferences. They define what must happen when an observed condition occurs. Appendix A expands these rules into pseudocode. Appendix C maps each rule to audit evidence and test vectors.

Implementation Model

A deployable EIT implementation requires five infrastructure components. The first is a consent registry that stores scope identifiers, authorized purposes, expiration dates, user-visible descriptions, revocation status, and jurisdictional constraints. The second is a logging service that creates LIL records only after consent validation. The third is a trajectory evaluation service that evaluates authorized windows of LIL records against deterministic TAR rules. The fourth is a response service that selects PDEV actions from a bounded response taxonomy. The fifth is an audit service that records the rule, evidence, decision, and integrity hash for each governance action.

This implementation model can operate internally without exposing a public API, but the functions must exist. A platform may implement EIT as middleware between the application interface and model orchestration layer, as a governance module inside a learning platform, or as an enterprise compliance service connected to deployed AI assistants. In each case, the system must prevent unauthorized longitudinal processing before it occurs rather than merely flagging it later.

The model is intentionally compatible with multiple deployment environments. A Class I implementation may provide logging and audit only. A Class II implementation may add trajectory monitoring without automated intervention. A Class III implementation may provide full logging, trajectory analysis, PDEV response selection, and audit export. This conformance tiering allows institutions to adopt the framework incrementally while preserving a clear path toward full governance capacity.

Conformance and Testability

Conformance class	Required capabilities	Required validation tests
Class I	Consent validation, LIL logging, audit records, retention enforcement.	Schema validation for LIL and audit records; consent bypass test; hash-chain integrity test.
Class II	All Class I controls plus TAR signal generation over authorized event windows.	Trajectory window test; no diagnostic inference test; severity classification test.
Class III	All Class II controls plus PDEV response selection, user disclosure, human review, and export.	Severity-to-response binding test; user-facing disclosure test; critical escalation test; record export test.

Conformance testing is central to EIT's legitimacy. The system should not claim compliance because it has adopted the vocabulary. It should claim compliance only when it can produce valid records, pass deterministic test vectors, preserve purpose boundaries, and demonstrate reviewable audit evidence.

Audit and Oversight Model

EIT auditability requires traceability across the full governance chain. A reviewer must be able to move from a user-facing response back to the TAR signal that triggered it, from the TAR signal back to the LIL records that supplied the evidence, from those records back to the consent scope that authorized processing, and from each step to the policy rule that governed the decision. This chain is the audit spine of the framework.

Oversight should be separated by role. System operators maintain the infrastructure. Compliance owners review rule execution and exception records. Independent auditors or institutional review bodies review conformance evidence and high-severity cases. Users or their authorized representatives should have access to readable records when those records affect user-facing treatment, access, or escalation. The audit system must therefore support both machine-readable validation and human-readable explanation.

Audit records must be generated for ordinary operations and exceptions. Ordinary audit records prove that the system followed the expected rule path. Exception audit records prove that the system detected a boundary failure, blocked or corrected the action, and routed the matter to the appropriate owner. A system with only exception logging cannot prove normal compliance. A system with only ordinary logging cannot prove that boundary failures were handled.

Limitation Clarity

EIT has a deliberately narrow scope. It governs longitudinal interaction telemetry, consent-bounded trajectory evaluation, response selection, and audit traceability at the application and platform interaction layer. It does not govern model training data, foundation model pretraining, compute allocation, model weight custody, frontier capability thresholds, or supply-chain security. Those domains require separate controls and should not be collapsed into EIT.

EIT also does not authorize psychological inference. The framework uses the term trajectory to describe observable governance patterns, such as repeated unresolved errors, consent boundary events, hidden adaptation risk, contestation events, or escalation failures. It does not classify a user as emotionally unstable, vulnerable, traumatized, impaired, or in need of therapeutic intervention. If an implementation uses EIT terminology to perform diagnostic or behavioral targeting, that implementation violates the framework.

A further limitation is that this paper is conceptual and standards-oriented. It provides deterministic logic, schema examples, and conformance tests, but it does not report empirical validation across deployed systems. Future work should test EIT implementations in controlled environments, evaluate usability of user-facing disclosures, assess audit burden, and compare outcomes across institutional contexts.

Discussion

The central value of EIT is that it moves governance from statement to operation. AI systems increasingly mediate education, work, support, search, and decision pathways. When these systems adapt across time, governance must follow the same temporal structure. One-time notice, static model documentation, and after-the-fact complaint procedures are insufficient when the system's behavior depends on accumulated interaction context.

EIT does not solve all AI governance problems. Its value is architectural specificity. It identifies a missing layer between high-level governance principles and deployed interaction behavior. It then defines the logic by which that layer can be implemented: consent before telemetry, observable records before trajectory signals, bounded signals before intervention, and audit evidence for every decision path.

This structure also reduces institutional ambiguity. A platform operator can identify what services must be built. A regulator can identify what records to request. A standards body can identify which sections should be normative. A civil society auditor can identify where user agency may fail. A researcher can identify what empirical studies are needed next. The framework therefore operates as a policy translation layer across technical, legal, institutional, and human-impact domains.

Conclusion

AI governance becomes real when it constrains system behavior. Emotional Infrastructure Telemetry provides a protocol-level framework for governing longitudinal AI interaction through consent boundaries, interaction logs, trajectory signals, response selection, and audit traceability. The paper's contribution is not a new ethical slogan. It is a mechanism for converting selected governance obligations into deterministic system behavior.

The framework is most useful where AI systems maintain interaction history and use that history to shape future responses. In those environments, user agency depends on more than model accuracy. It depends on whether the system has authority to remember, whether the user can understand how prior context is being used, whether harmful trajectories can be detected without profiling the user, whether proportionate responses exist, and whether an auditor can reconstruct the chain of authority. EIT gives those questions a system architecture.

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Appendix A

Deterministic Protocol Logic

This appendix provides protocol-level pseudocode for the EIT policy translation layer.

The logic is intentionally deterministic: the same valid input state should produce the same governance decision, audit record, and escalation pathway.

```
function process_interaction(event, consent_registry, policy_rules):
    consent_scope = consent_registry.lookup(event.user_id, event.requested_purpose)

    if consent_scope is null or consent_scope.status != "active":
        audit("EIT-R1", event.id, "blocked_no_valid_consent")
        return deny_longitudinal_processing(event)

    if event.requested_purpose not in consent_scope.authorized_purposes:
        audit("EIT-R2", event.id, "blocked_purpose_mismatch")
        escalate("compliance_review", event.id)
        return deny_longitudinal_processing(event)

    lil_record = create_lil_record(event, consent_scope)
    audit("EIT-R3", lil_record.record_id, "lil_record_created")

    if event.requires_trajectory_evaluation == false:
        return continue_without_tar(lil_record)

    authorized_window = get_authorized_lil_window(
        user_id = event.user_id,
        consent_scope_id = consent_scope.id,
        start = consent_scope.window_start,
        end = now()
    )

    tar_candidate = evaluate_tar_rules(authorized_window)

    if tar_candidate.contains_prohibited_inference == true:
        audit("EIT-R4", tar_candidate.id,
            "blocked_diagnostic_or_vulnerability_inference")
        escalate("policy_review", tar_candidate.id)
        return continue_without_tar(lil_record)

    tar_signal = create_tar_signal(tar_candidate)
    audit("EIT-R5", tar_signal.signal_id,
        "tar_signal_created_with_authorized_window")

    response = select_pdev_response(tar_signal.severity, tar_signal.signal_type)

    if tar_signal.severity in ["high", "critical"]:
        response.requires_human_review = true
        audit("EIT-R6", response.response_id, "severity_bound_response_selected")
        escalate("human_review", response.response_id)

    if response.changes_user_experience == true:
        response.user_visible_notice = generate_disclosure(tar_signal)
        audit("EIT-R7", response.response_id, "adaptive_disclosure_generated")

    audit("EIT-R8", [lil_record.record_id, tar_signal.signal_id,
        response.response_id], "integrity_chain_updated")
    return deliver_response(response)
```

Appendix B

Canonical Schema Examples

The package includes machine-readable JSON schema files for LIL, TAR, PDEV, and audit records. The excerpts below show the field structure expected in a protocol-level manuscript. The schema files bundled with this submission package provide the complete machine-readable versions.

lil-event-v1.json

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://emotionalinfrastructure.org/schemas/eit/lil-event-v1.json",
  "title": "LIL Event Record",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "record_id",
    "schema_version",
    "timestamp_utc",
    "actor_type",
    "system_context",
    "consent_scope_id",
    "interaction_type",
    "event_summary",
    "event_hash"
  ],
  "properties": {
    "record_id": {
      "type": "string",
      "description": "Stable unique identifier for the event record."
    },
    "schema_version": {
      "type": "string",
      "const": "lil-event-v1"
    },
    "timestamp_utc": {
      "type": "string",
      "format": "date-time"
    },
    "actor_type": {
      "type": "string",
      "enum": [
        "user",
        "system",
        "operator",
        "third_party"
      ]
    },
    "system_context": {
      "type": "object",
      "required": [
        "application_id",
        "model_or_service_id",
        "session_id"
      ]
    }
  }
}
```



```

    ],
    "additionalProperties": false,
    "properties": {
      "application_id": {
        "type": "string"
      },
      "model_or_service_id": {
        "type": "string"
      },
      "session_id": {
        "type": "string"
      }
    }
  },
  "consent_scope_id": {
    "type": "string"
  },
  "interaction_type": {
    "type": "string",
    "enum": [
      "input",
      "output",
      "correction",
      "refusal",
      "escalation",
      "disclosure",
      "user_feedback"
    ]
  },
  "event_summary": {
    "type": "string",
    "maxLength": 1000
  },
  "observable_effect": {
    "type": "string",
    "enum": [
      "none",
      "confusion",
      "repetition",
      "task_blockage",
      "safety_escalation",
      "policy_boundary",
      "user_contestation"
    ]
  },
  "excluded_inference": {
    "type": "boolean",
    "description": "True indicates that prohibited diagnostic or psychological inference was not performed."
  },
  "event_hash": {
    "type": "string"
  },
  "previous_event_hash": {
    "type": [
      "string",
      "null"
    ]
  }
}
}

```

tar-signal-v1.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",

```

```

"$id": "https://emotionalinfrastructure.org/schemas/eit/tar-signal-v1.json",
"title": "TAR Trajectory Signal",
"type": "object",
"additionalProperties": false,
"required": [
  "signal_id",
  "schema_version",
  "timestamp_utc",
  "consent_scope_id",
  "source_window",
  "signal_type",
  "severity",
  "confidence_basis",
  "recommended_response",
  "audit_record_id"
],
"properties": {
  "signal_id": {
    "type": "string"
  },
  "schema_version": {
    "type": "string",
    "const": "tar-signal-v1"
  },
  "timestamp_utc": {
    "type": "string",
    "format": "date-time"
  },
  "consent_scope_id": {
    "type": "string"
  },
  "source_window": {
    "type": "object",
    "required": [
      "start_timestamp_utc",
      "end_timestamp_utc",
      "record_count"
    ],
    "properties": {
      "start_timestamp_utc": {
        "type": "string",
        "format": "date-time"
      },
      "end_timestamp_utc": {
        "type": "string",
        "format": "date-time"
      },
      "record_count": {
        "type": "integer",
        "minimum": 1
      }
    }
  },
  "signal_type": {
    "type": "string",
    "enum": [
      "continuity_disruption",
      "consent_boundary_risk",
      "user_contestation",
      "unresolved_error_pattern",
      "intervention_needed"
    ]
  },
  "severity": {
    "type": "string",
    "enum": [
      "low",

```

```

        "medium",
        "high",
        "critical"
    ]
},
"confidence_basis": {
    "type": "array",
    "items": {
        "type": "string"
    },
    "minItems": 1
},
"recommended_response": {
    "type": "string",
    "enum": [
        "no_action",
        "disclose",
        "clarify",
        "pause",
        "human_review",
        "export_record"
    ]
},
"audit_record_id": {
    "type": "string"
}
}
}

```

pdev-response-v1.json

```

{
    "$schema": "https://json-schema.org/draft/2020-12/schema",
    "$id": "https://emotionalinfrastructure.org/schemas/eit/pdev-response-v1.json",
    "title": "PDEV Intervention Response",
    "type": "object",
    "additionalProperties": false,
    "required": [
        "response_id",
        "schema_version",
        "timestamp_utc",
        "linked_signal_id",
        "response_type",
        "agency_preservation_step",
        "user_visible_notice",
        "audit_record_id"
    ],
    "properties": {
        "response_id": {
            "type": "string"
        },
        "schema_version": {
            "type": "string",
            "const": "pdev-response-v1"
        },
        "timestamp_utc": {
            "type": "string",
            "format": "date-time"
        },
        "linked_signal_id": {
            "type": "string"
        },
        "response_type": {
            "type": "string",
            "enum": [
                "disclosure",

```

```

        "clarification",
        "safe_pause",
        "human_review",
        "record_export",
        "policy_refusal"
    ]
},
"agency_preservation_step": {
    "type": "string",
    "enum": [
        "user_choice_prompt",
        "plain_language_explanation",
        "appeal_route",
        "session_boundary",
        "data_access_route"
    ]
},
"user_visible_notice": {
    "type": "string",
    "maxLength": 1200
},
"operator_action_required": {
    "type": "boolean"
},
"audit_record_id": {
    "type": "string"
}
}
}

```

audit-record-v1.json

```

{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://emotionalinfrastructure.org/schemas/eit/audit-record-v1.json",
  "title": "EIT Audit Record",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "audit_record_id",
    "schema_version",
    "timestamp_utc",
    "linked_records",
    "policy_rule_id",
    "decision_outcome",
    "decision_basis",
    "integrity_hash"
  ],
  "properties": {
    "audit_record_id": {
      "type": "string"
    },
    "schema_version": {
      "type": "string",
      "const": "audit-record-v1"
    },
    "timestamp_utc": {
      "type": "string",
      "format": "date-time"
    },
    "linked_records": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "minItems": 1
    }
  }
}

```

```

    },
    "policy_rule_id": {
      "type": "string"
    },
    "decision_outcome": {
      "type": "string"
    },
    "decision_basis": {
      "type": "array",
      "items": {
        "type": "string"
      }
    },
    "review_owner": {
      "type": "string"
    },
    "retention_class": {
      "type": "string",
      "enum": [
        "short",
        "standard",
        "extended_by_policy"
      ]
    },
    "integrity_hash": {
      "type": "string"
    },
    "previous_integrity_hash": {
      "type": [
        "string",
        "null"
      ]
    }
  }
}

```

Appendix C

Audit Traceability Matrix

Rule	Trigger condition	Control action	Required audit evidence	Test vector
EIT-R1	Consent scope is missing or inactive.	Block longitudinal processing.	consent_scope_id, audit_record_id, decision_outcome	TV-01
EIT-R2	Purpose is outside authorized scope.	Deny processing and escalate.	requested_purpose, authorized_purposes, policy_rule_id	TV-02
EIT-R3	Governance-relevant event occurs.	Create LIL record.	record_id, timestamp_utc, event_hash	TV-03
EIT-R4	Prohibited diagnostic inference attempted.	Reject TAR signal.	signal_type, decision_basis, prohibited_inference_flag	TV-04
EIT-R5	Trajectory evaluation requested.	Restrict analysis to authorized window.	source_window, record_count, consent_scope_id	TV-05

Rule	Trigger condition	Control action	Required audit evidence	Test vector
EIT-R6	High or critical severity signal exists.	Bind severity to required PDEV response.	severity, response_type, linked_signal_id	TV-06
EIT-R7	Behavior changes using prior context.	Generate user-facing disclosure.	user_visible_notice, response_id, audit_record_id	TV-07
EIT-R8	Governance record is created.	Update audit hash chain.	integrity_hash, previous_integrity_hash	TV-08
EIT-R9	Telemetry requested for prohibited secondary use.	Deny request and escalate.	requested_use, decision_outcome, review_owner	TV-09
EIT-R10	Authorized record access request occurs.	Export readable and machine-readable record.	export_id, linked_records, access_basis	TV-10
EIT-R11	Retention period expires.	Delete or archive according to policy.	retention_class, timestamp_utc, deletion_status	TV-11
EIT-R12	Conformance claim is made.	Validate class requirements.	conformance_class, test_results, validation_status	TV-12

Appendix D

Test Vectors

Test vector	Scenario	Input condition	Expected output
TV-01	No active consent scope.	Interaction event requests longitudinal processing.	Processing denied; consent exception audit created.
TV-02	Consent allows continuity support only.	Telemetry requested for behavioral ranking.	Processing denied; prohibited purpose audit created.
TV-03	Valid consent and governance event.	System output creates user contestation event.	LIL record created with event hash and consent scope.
TV-04	Authorized LIL window exists.	TAR rule attempts to label user vulnerability.	Signal rejected; policy review escalation created.
TV-05	Valid consent window is seven days.	TAR attempts to include older events.	Older records excluded; boundary reduction audited.
TV-06	High severity unresolved error pattern.	Three unresolved task blockages within authorized window.	PDEV response selects disclosure and human review.
TV-07	System changes answer strategy	User receives adapted response.	Plain-language adaptive disclosure

Test vector	Scenario	Input condition	Expected output
	using prior context.		shown.
TV-08	Audit record created.	Record hash chain is updated.	Integrity hash and previous hash are present.
TV-09	Internal service requests EIT records for advertising.	Request enters telemetry service.	Request denied; compliance review triggered.
TV-10	User requests governance record export.	Records affect user-facing treatment.	Readable and machine-readable export generated.

Appendix E

Data Minimization and Non-Use Clause

EIT records must be limited to governance-relevant evidence. Systems should avoid storing full interaction content when a bounded event summary, event type, timestamp, consent scope identifier, and integrity hash are sufficient for traceability. When full content is required for dispute resolution or institutional review, the record must be assigned a retention class and access control level.

EIT telemetry must not be used for advertising, commercial ranking, behavioral targeting, psychological profiling, manipulation optimization, or unrelated personalization. This clause is not optional. It preserves the distinction between governance telemetry and exploitation telemetry. An implementation that uses EIT records for secondary targeting should not be described as EIT-compliant.

Appendix F

Submission Readiness Checklist

Requirement	Status
Pipeline figure included	Complete
References expanded to 20-30 sources	Complete

Requirement	Status
Internal algorithmic auditing literature integrated	Complete
Socio-technical infrastructure literature integrated	Complete
Limitation section clarifies non-coverage of training data and compute-level risk	Complete
Pseudocode included for protocol-level determinism	Complete
Schema examples included and bundled as JSON files	Complete
Audit traceability matrix included	Complete
Test vectors included	Complete
No unresolved placeholders	Complete